

Counterfactual Explanations of AI Systems

Fundamentals, Methods, & User Studies for XAI

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The Problem

ChatGPT OpenAI



<https://youtu.be/eMx-2s7mZ24>



AI is out there...being deployed...busily making mistakes !

Spot the Ball !



Spot the Head !

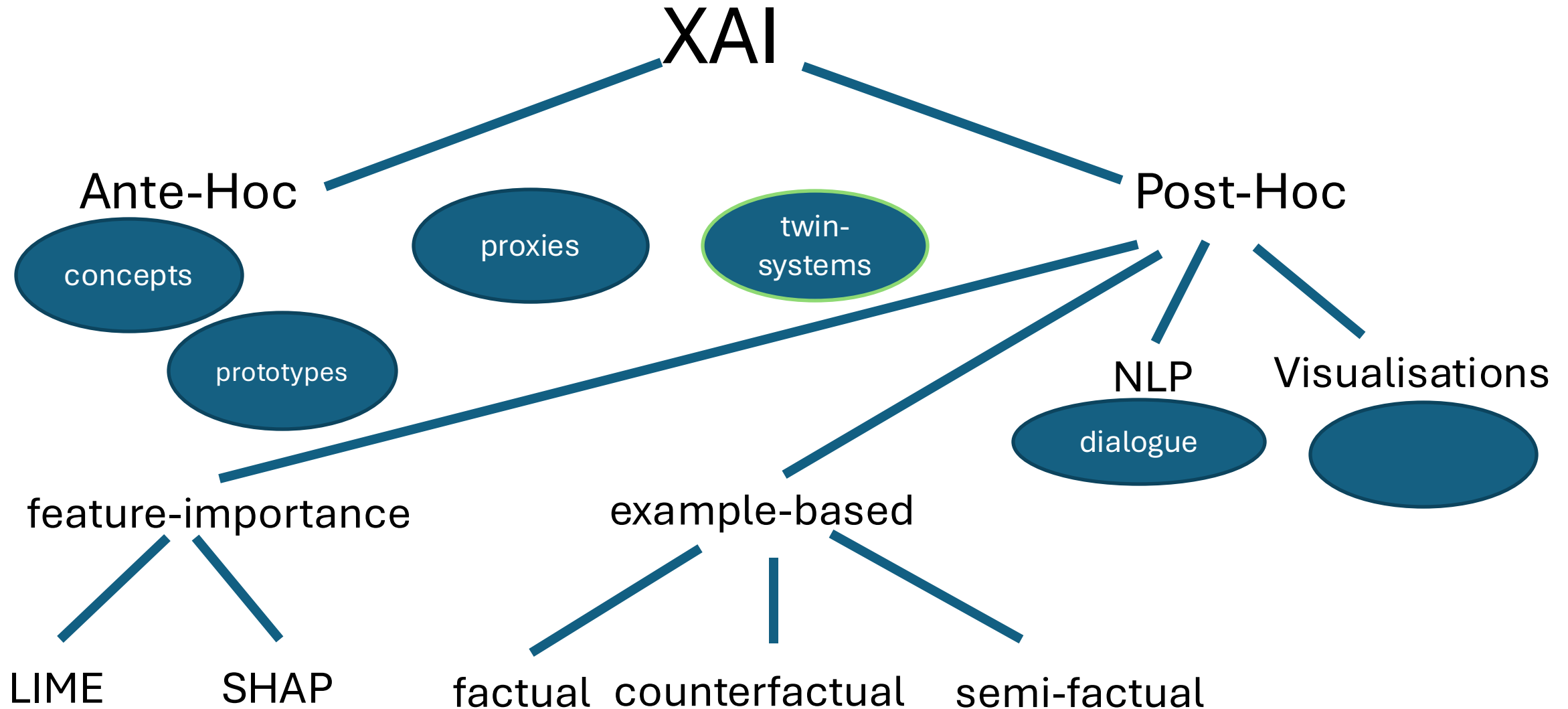


One Solution: Explain It !

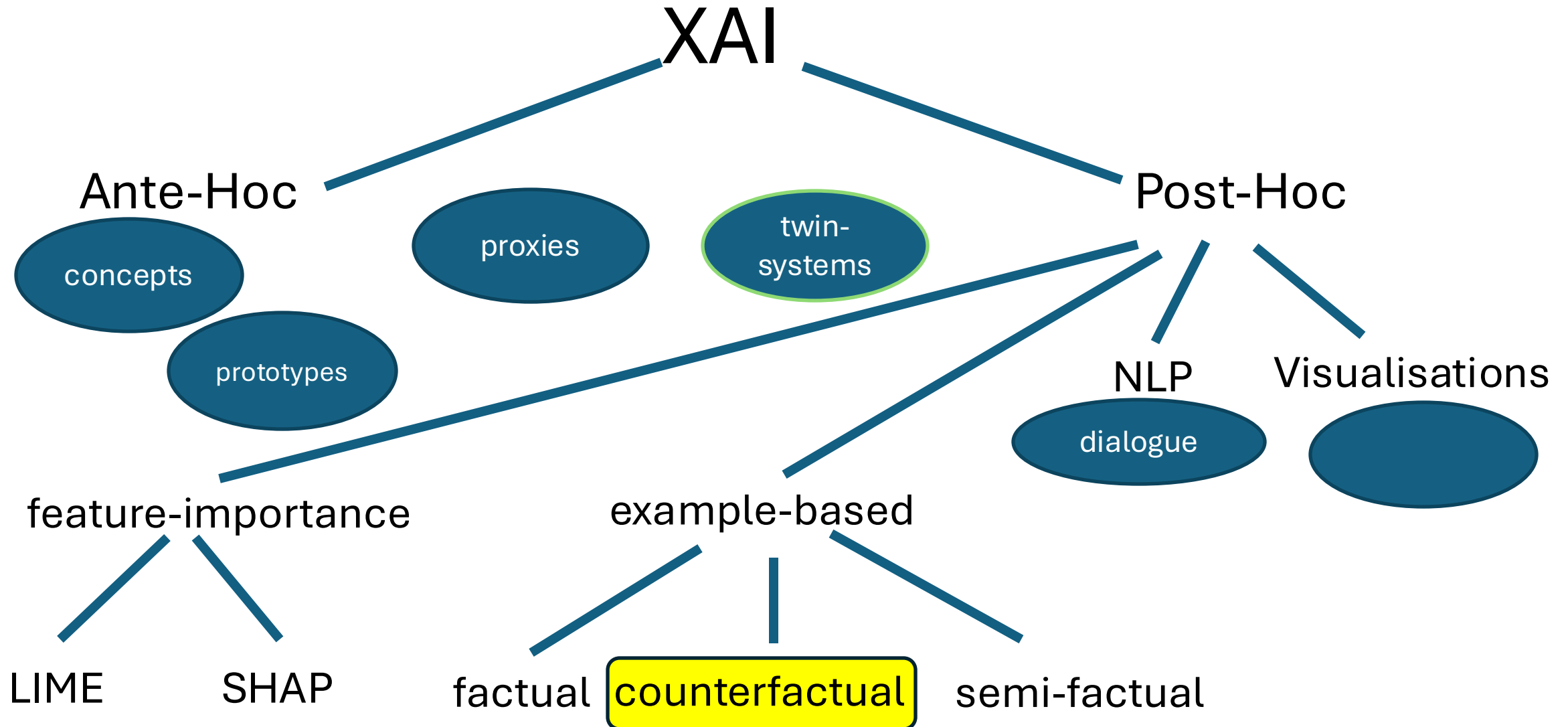
- Ante-Hoc: Somehow show the workings of the machine (*simulatability*)
- Post-Hoc: Justification after-the-fact to *explain* workings (*evidence of what happened*)



Explainable AI: Taxonomy



Explainable AI: Taxonomy



Counterfactual Explanations

***Your loan
application has
been refused !***

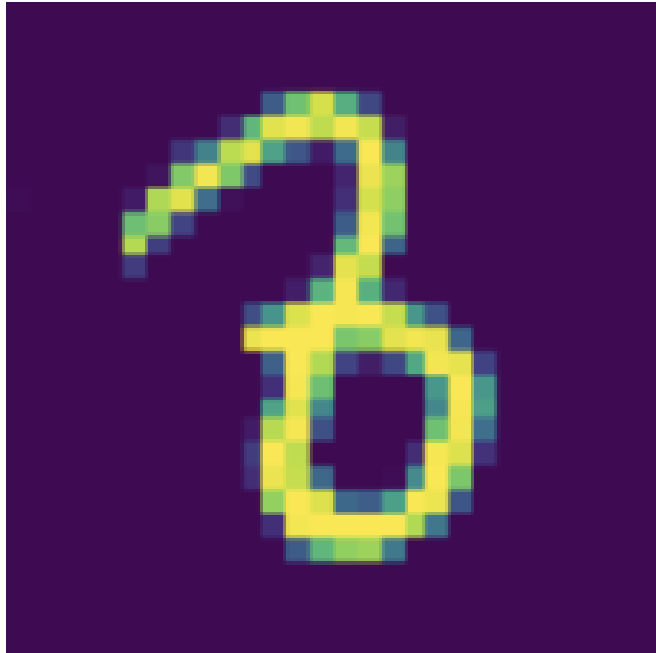
***If you earned 5k
more, you would have
gotten the loan.***

XAI



Counterfactual Explanations

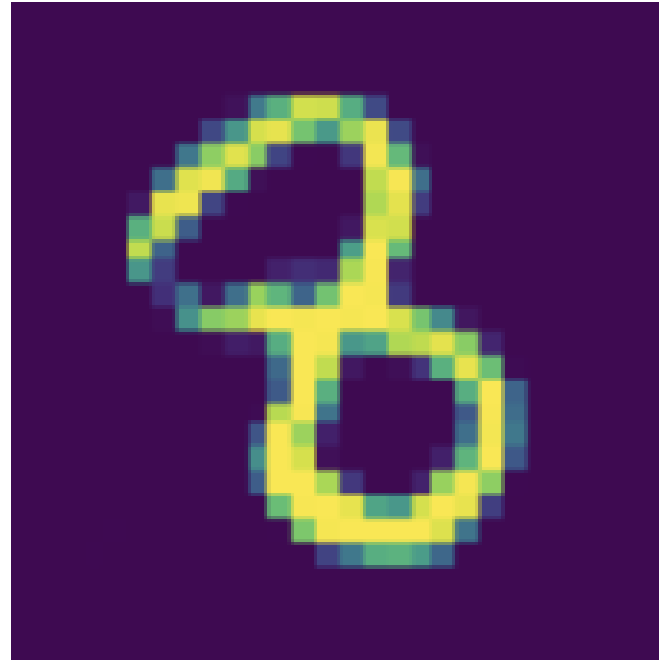
If the test image looked like this, I would have thought it was an “8”.



Test Image

Label: 8

Prediction: 3

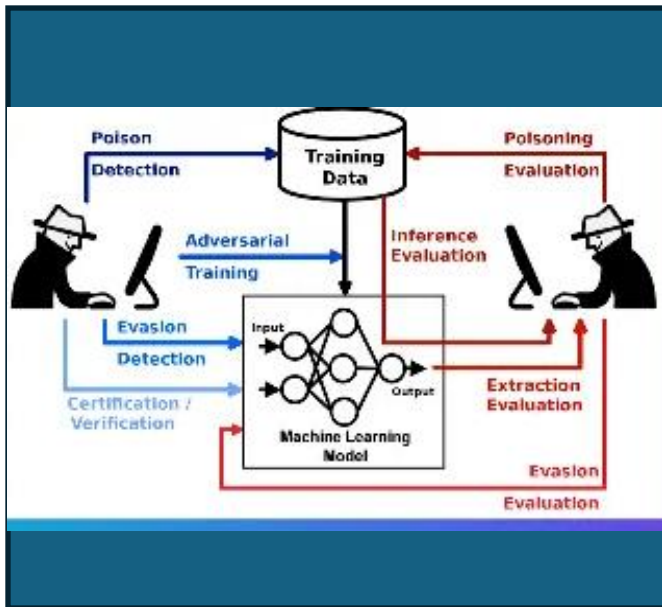


Counterfactual

New Prediction: 8

Why Counterfactuals?

Artificial Intelligence



Looks like adversarial learning...

[Wachter et al \(2019\) Facct-19](#)

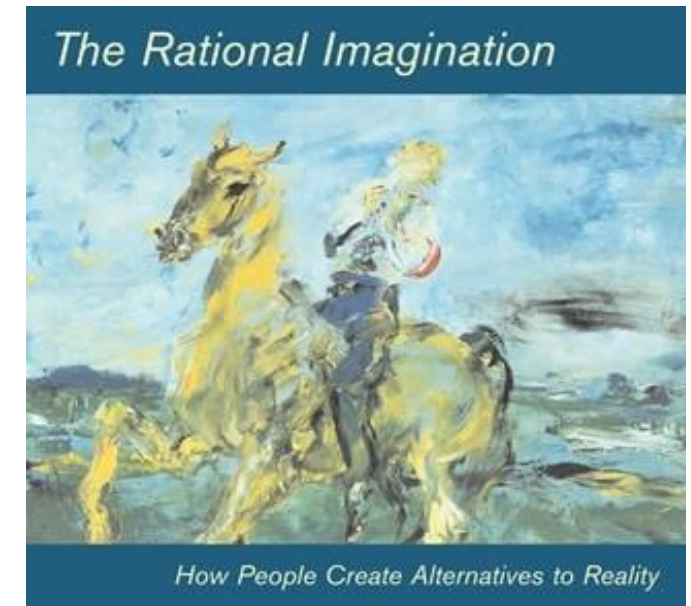
Legal



GDPR compliant...

[Wachter et al \(2017\) Harv. JL &B Tech.](#)

Psychology



Used by people spontaneously...

[Byrne \(2007\) MIT Press](#)

Plan for the Day...

Time	Topics
8:30 AM	Introduction
	<i>Hello and Introductions!</i>
8:35 AM	From Philosophy of Counterfactuals to XAI and User Studies
	<i>Philosophical & Psychological Fundamentals of Counterfactuals</i>
	<i>Getting Into the User Study Mindset</i>
	Q&A
9:10 AM	Computational Fundamentals of Counterfactuals in AI
	<i>Formalisation</i>
	<i>Current and Future Directions</i>
	Q&A
10:00 AM	COFFEE (10:00-10:30)
10:30 AM	How to Run a User Study
	<i>A Roadmap: From Motivation to Results</i>
	Q&A
11:15 AM	Hands-on
	Hands-on: <i>Computing Counterfactuals</i>
	Hands-on: <i>Analyzing a User Study</i>
12:20 PM	Closing Session, Discussion and Final Q&A
TUTORIAL END	

